

Redis SQL Trino

Version 0.4.0

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Introduction

Redis SQL Trino is a [Trino](#) connector which allows access to Redisearch data from Trino.

This guide provides documentation and usage information across the following topics:

- [Installation](#)
- [Configuration](#)
- [SQL support](#)
- [Clients](#)
- [Walkthrough](#)

Installation

Trino

Trino installation instructions are available at <https://trino.io/docs/current/installation.html>.

RediSearch connector

Download [latest release](#) and unzip without any directory structure under:

`<trino>/plugin/redisearch`

Create a RediSearch connector configuration file and change/add [properties](#) as needed.

Configuration

To configure the Redisearch connector, create a catalog properties file and change/add properties as needed.

etc/catalog/redisearch.properties

```
connector.name=redisearch
redisearch.uri=redis://localhost:6379
```

Table 1. Connector properties

Property name	Description	Default
<code>redisearch.default-schema-name</code>	The schema that contains all tables defined without a qualifying schema name.	<code>default</code>
<code>redisearch.case-insensitive-names</code>	Match index names case insensitively.	<code>false</code>
<code>redisearch.cursor-count</code>	Number of rows read during each aggregation cursor fetch.	<code>1000</code>
<code>redisearch.table-cache-refresh</code>	Seconds an index's schema is cached. Indexes and fields created outside Trino can take this long to appear.	<code>60</code>

Table 2. Redis connection properties

Property name	Description	Default
<code>redisearch.uri</code>	A Redis connection string. Redis URI syntax .	
<code>redisearch.username</code>	Redis connection username.	
<code>redisearch.password</code>	Redis connection password.	
<code>redisearch.cluster</code>	Connect to a Redis Cluster.	<code>false</code>
<code>redisearch.resp2</code>	Force Redis protocol version to RESP2.	<code>false</code>

The Redisearch connector provides additional security options to support Redis servers with TLS mode.

Table 3. TLS properties

Property name	Description	Default
<code>redisearch.insecure</code>	Allow insecure connections (e.g. invalid certificates) when using SSL.	<code>false</code>

Property name	Description	Default
<code>redisearch.cacert-path</code>	X.509 CA certificate file to verify with.	
<code>redisearch.key-path</code>	PKCS#8 private key file to authenticate with (PEM format).	
<code>redisearch.key-password</code>	Password of the private key file, or null if it's not password-protected.	
<code>redisearch.cert-path</code>	X.509 certificate chain file to authenticate with (PEM format).	

SQL support

Tables and columns

Each Redis Query Engine index is a table in the `default` schema (see `redisearch.default-schema-name`). A table has these columns:

- One per indexed field, named by its attribute (the `AS` name for JSON paths).
- One per field that isn't indexed but appears in the first 10 documents returned by `FT.SEARCH <index> *`.
- For JSON indexes, a `$` column holding the whole document as JSON text.
- A hidden `key` column holding the document's Redis key. Hidden columns are left out of `SELECT *, so name it: SELECT key, * FROM beers`.

Table 4. Type mapping

Redis field type	Trino type
NUMERIC	DOUBLE
TAG, TEXT, GEO, GEOSHAPE, VECTOR	VARCHAR
Not indexed	VARCHAR

After `FT.CREATE` over existing keys, or `ALTER TABLE ... ADD COLUMN` (which runs `FT.ALTER`), Redis indexes the documents in the background. Until it finishes, queries on that index fail with `RESEARCH_INDEX_NOT_READY` (Index `beers` is still being built (42% indexed) ...) instead of returning incomplete results. Retry once `FT.INFO` reports `indexing` as `0`. `CREATE TABLE` creates its index with `SKIPINITIALSCAN`, so a new table can be queried right away.

Statements

Table 5. Supported statements

Statement	Notes
<code>SELECT</code>	All of Trino's query syntax, including joins (also across catalogs), subqueries, <code>WITH</code> , <code>UNION</code> , <code>DISTINCT</code> , <code>GROUP BY</code> , <code>ORDER BY</code> and window functions. Pushdown describes which parts run in Redis.
<code>SHOW SCHEMAS</code> , <code>SHOW TABLES</code> , <code>SHOW COLUMNS</code> , <code>DESCRIBE</code> , <code>SHOW CREATE TABLE</code> , <code>information_schema</code>	<code>SHOW TABLES</code> lists the indexes returned by <code>FT._LIST</code> .
<code>EXPLAIN</code>	The table scan shows the filters, limit and aggregations sent to Redis.
<code>CREATE TABLE</code>	Creates a hash index named after the table, with key prefix <code><table>:</code> . See column types . <code>IF NOT EXISTS</code> is supported.

Statement	Notes
CREATE TABLE AS SELECT	Columns get the same field types as in CREATE TABLE. A query with a column of another type fails before the table is created.
INSERT, INSERT ... SELECT	Hash indexes only: the connector writes hashes, which an index on JSON documents never sees, so it fails on a JSON index. Each row is a new hash with key <code><prefix><ULID></code> , where <code><prefix></code> is the index's first key prefix.
UPDATE	Hash indexes only. Setting a column to NULL removes the field from the hash.
DELETE	Deletes the matching keys, on hash and JSON indexes.
MERGE	Hash indexes only, except that WHEN MATCHED THEN DELETE also works on JSON indexes.
ALTER TABLE ... ADD COLUMN	Adds a field to the index with FT.ALTER. Columns are always added at the end. Queries on the table fail until Redis finishes reindexing its documents.
DROP TABLE	Drops the index and deletes its documents (FT.DROPINDEX ... DD).

These statements fail with a `This connector does not support ...` error:

- CREATE SCHEMA, DROP SCHEMA, ALTER SCHEMA
- CREATE OR REPLACE TABLE
- ALTER TABLE ... RENAME TO, DROP COLUMN, RENAME COLUMN, ALTER COLUMN ... SET DATA TYPE
- TRUNCATE
- COMMENT ON
- CREATE VIEW, CREATE MATERIALIZED VIEW
- Queries on table versions (FOR VERSION AS OF, FOR TIMESTAMP AS OF)

Writes also fail when fault-tolerant execution is enabled (`retry-policy` other than `NONE`).

Table 6. Column types in CREATE TABLE

Trino type	Index field	Read back as
BIGINT, INTEGER, SMALLINT, TINYINT, DOUBLE, REAL, DECIMAL	NUMERIC	DOUBLE
TIMESTAMP(3), TIMESTAMP(3) WITH TIME ZONE	NUMERIC	DOUBLE, in epoch milliseconds
VARCHAR, CHAR, UUID	TAG	VARCHAR
BOOLEAN, DATE	TAG	VARCHAR: <code>true</code> or <code>false</code> , and ISO dates such as <code>2024-01-02</code>

Other types, such as ARRAY, MAP, ROW, VARBINARY and TIME, fail with `Unsupported column type`.

Because columns are read back from the index, a column declared as BIGINT becomes DOUBLE and one declared as BOOLEAN or DATE becomes VARCHAR, so later **INSERT** statements must supply values of those types. The TAG fields aren't full-text TEXT fields. They're **CASESENSITIVE** and split values into tags on the ASCII unit separator (`\x1f`) instead of `,`, so Redis can prefilter `=` and **IN** on values containing commas, and doesn't return rows that only differ in case.

Pushdown

Redis evaluates these parts of a query. Trino evaluates everything else on the rows Redis returns.

- Filters on indexed fields, combined with **AND**:
 - `=`, **IN**, `<`, `<=`, `>`, `>=` and **BETWEEN** on NUMERIC fields.
 - `=` and **IN** on TAG fields, as a tag query. Redis splits stored values into tags on the field's **SEPARATOR** (by default `,` on hashes and none on JSON), trims whitespace around each tag, and ignores case unless the field is **CASESENSITIVE**. It returns the documents with a tag that matches the value, and Trino keeps those equal to it. Trino evaluates the filter alone when a value contains the separator, starts or ends with whitespace, or has a control character other than whitespace, since a tag query can't match it.
 - `=` and **IN** on TEXT fields, as a full-text query for the value's words. Redis returns the documents containing all of them, and Trino keeps those equal to the value. Trino evaluates the filter alone when a value has only stop words, and for every TEXT field of an index created with its own **STOPWORDS**.

Redis can't match a missing field or compare strings, so Trino evaluates **IS NULL**, **IS NOT NULL**, a condition combined with **OR ... IS NULL**, and `<>`, `<`, `>` and **BETWEEN** on VARCHAR columns.

Trino also evaluates **LIKE**. A Redis wildcard query matches single words of a TEXT field, ignores case on a TAG field without **CASESENSITIVE**, and leaves out documents once it has matched **MAXEXPANSIONS** (200 by default) distinct values.

- `count()`, `min`, `max`, `sum` and `avg` on numeric columns, with or without **GROUP BY**, as **FT.AGGREGATE** with **GROUPBY** and **REDUCE**. If any aggregate in a query can't be pushed down, such as `count(DISTINCT ...)` or `count(column)`, or Trino filters the rows itself, as it does for filters on TAG and TEXT fields, Trino computes all of them. `count(column)` skips nulls, but Redis's **COUNT** counts documents, so only `count()` is pushed down. Trino evaluates **HAVING** on the groups Redis returns.
- **LIMIT**, unless Trino filters the rows itself. Trino sorts **ORDER BY ... LIMIT** itself.

Known limitations

Queries known to return wrong results or fail are tracked as [open issues labeled bug](#).

Clients

JDBC Driver

The Trino JDBC driver allows users to access Trino from Java-based applications, and other non-Java applications running in a JVM.

Refer to the [Trino documentation](#) for setup instructions.

The following is an example of a JDBC URL used to create a connection to Redis SQL Trino:

```
jdbc:trino://example.net:8080/redisearch/default
```

Tableau

Refer to the [Tableau documentation](#) for setup instructions.

Trino CLI

Refer to the [Trino CLI documentation](#) for setup instructions.

Build

Run these commands to build the Trino connector for Redisearch from source (requires Java 25+):

```
git clone https://github.com/redis-field-engineering/redis-sql-trino.git
cd Redis SQL Trino
./mvnw clean package -DskipTests
```

Complete Walkthrough

Follow these step-by-step instructions to deploy a single-node Trino server on Ubuntu.

Install Java

Trino requires a 64-bit version of Java 25. It is recommended to use [Azul Zulu](#) as the JDK.

```
$ java -version
openjdk version "25.0.1" 2025-10-21 LTS
OpenJDK Runtime Environment Zulu25.30+17-CA (build 25.0.1+8-LTS)
OpenJDK 64-Bit Server VM Zulu25.30+17-CA (build 25.0.1+8-LTS, mixed mode, sharing)
```

Set up Trino

Download the Trino server tarball and unpack it.

```
wget https://repo1.maven.org/maven2/io/trino/trino-server/483/trino-server-483.tar.gz
mkdir /usr/lib/trino
tar xzvf trino-server-483.tar.gz --directory /usr/lib/trino --strip-components 1
```

Trino needs a data directory for storing logs, etc. It is recommended to create a data directory outside of the installation directory, which allows it to be easily preserved when upgrading Trino.

Create a data directory

```
mkdir -p /var/trino
```

Create an **etc** directory inside the installation directory to hold configuration files.

```
mkdir /usr/lib/trino/etc
```

Create a **node.properties** file.

/usr/lib/trino/etc/node.properties

```
node.environment=production
node.id=ffffffff-ffff-ffff-ffff-ffffffffffffff
node.data-dir=/var/trino
```

Create a JVM config file.

/usr/lib/trino/etc/jvm.config

```
-server
-Xmx16G
-XX:InitialRAMPercentage=80
-XX:MaxRAMPercentage=80
-XX:G1HeapRegionSize=32M
-XX:+ExplicitGCInvokesConcurrent
-XX:+ExitOnOutOfMemoryError
-XX:+HeapDumpOnOutOfMemoryError
-XX:-OmitStackTraceInFastThrow
-XX:ReservedCodeCacheSize=512M
-XX:PerMethodRecompilationCutoff=10000
-XX:PerBytecodeRecompilationCutoff=10000
-Djdk.attach.allowAttachSelf=true
-Djdk.nio.maxCachedBufferSize=2000000
--add-modules=jdk.incubator.vector
```

Create a config properties file.

/usr/lib/trino/etc/config.properties

```
coordinator=true
node-scheduler.include-coordinator=true
http-server.http.port=8080
discovery.uri=http://localhost:8080
```

Create a logging configuration file.

/usr/lib/trino/etc/log.properties

```
io.trino=INFO
```

Set up Redis SQL Trino

Download latest [release](#) and unzip without any directory structure under **plugin/redisearch**:

```
wget https://github.com/redis-field-engineering/redis-sql-trino/releases/download/v0.4.0/redis-sql-trino-0.4.0.zip
unzip -j redis-sql-trino-0.4.0.zip -d /usr/lib/trino/plugin/redisearch
```

Create a etc/catalog subdirectory:

```
mkdir /usr/lib/trino/etc/catalog
```

Create a Redisearch connector configuration file:

/usr/lib/trino/etc/catalog/redisearch.properties

```
connector.name=redisearch  
redisearch.uri=redis://localhost:6379
```

Change and/or add [properties](#) as needed.

Start Trino

Start the Trino server:

```
/usr/lib/trino/bin/launcher run
```

Run Trino CLI

Download and run [trino-cli-483-executable.jar](#):

```
wget https://repo1.maven.org/maven2/io/trino/trino-cli/483/trino-cli-483-  
executable.jar -O /usr/local/bin/trino  
chmod +x /usr/local/bin/trino  
trino --catalog redisearch --schema default
```

Run a SQL query:

```
trino:default> select * from mySearchIndex;
```